

Predicting Symptomatic Knee Osteoarthritis Risk from CHARLS

Data:

Internal Validation, Biological Age Decomposition, and
Cross-Cultural Transportability Assessment on HRS and ELSA

BioAge-KOA-Prediction Team

July 2026

Abstract

Objective. To develop and internally validate a machine-learning model for symptomatic knee osteoarthritis (KOA) risk prediction in a CHARLS-derived Chinese cohort, to decompose the contribution of biological age (BA) to model performance, and to characterise transportability to US (HRS) and UK (ELSA) aging cohorts.

Methods. A Random Forest model ($n = 300$ trees) was trained on 17 sociodemographic and clinical variables (raw17) from CHARLS Dataset A ($n = 12,329$, KOA prevalence 13.3%). Internal validation used stratified 5-fold out-of-fold estimation and an early 80/20 stratified unseen holdout. To isolate the contribution of the pre-computed CHARLS “Biological Age” field — whose derivation method is undocumented — we (i) computed transparent alternatives using the Klemra–Doubal method (KDM-BA) and an adapted PhenoAge surrogate from 8 blood biomarkers, (ii) replaced Sheet1 BA with each alternative inside raw17 while holding feature count constant, and (iii) evaluated each BA variant as a single-feature KOA predictor. To test whether a demographic-predicted BA could transport the aging signal to cohorts lacking biomarkers, we trained ML regressors (Random Forest, XGBoost, LightGBM ensemble) to predict both Sheet1 BA and KDM-BA from raw16 features alone ($R^2 = 0.64$ and 0.48 , respectively), then applied the predicted BA to HRS Wave 13 ($n = 20,605$) and ELSA Wave 8 ($n = 8,416$) using harmonised raw16 feature sets.

Results. The primary model achieved ROC-AUC 0.847 (internal OOF) / 0.907 (unseen holdout). Replacing Sheet1 BA with KDM-BA degraded RF AUROC from 0.896 to 0.831 ($\Delta = -0.065$); removing BA entirely produced 0.859 ($\Delta = -0.037$). As a single feature, all BA variants achieved only 0.50–0.54 AUROC in CHARLS, barely above chance. ML-predicted BA applied to external cohorts improved over raw16 by only +0.003 to +0.010 AUROC. Restricting CHARLS to raw16 produced ROC-AUC 0.804 / 0.869 — a feature-subset penalty of $\Delta = -0.038$. Applying the same raw16 model to ELSA and HRS yielded ROC-AUC 0.604 for both cohorts — a cross-cultural penalty of $\Delta = -0.265$. The total AUC gap (-0.303) decomposes as 13% feature-subset vs. 87% cross-cultural. The RAS experiment recovered only $\Delta = +0.006$ AUC.

Conclusions. The biomarker–BA–KOA relationship is population-specific: BA is a weak standalone KOA predictor ($r = 0.02$ – 0.04) even in CHARLS, and demographic-predicted BA adds negligible signal to external cohorts. The dominant driver of performance degradation across populations is cross-cultural — outcome definition heterogeneity, diagnostic ascertainment, and population demographic differences — not the absence of Biological Age or the choice of BA computation method. These findings reframe the external validation challenge: the primary obstacle is not a missing feature but a non-transferable disease–aging relationship.

Keywords: symptomatic KOA, machine learning, biological age, KDM, cross-cultural transportability, TRIPOD, calibration, DCA, HRS, ELSA

1 Introduction

Knee osteoarthritis is one of the leading causes of years lived with disability worldwide, with age-standardised prevalence having risen steadily in recent decades[4]. In aging populations, symptomatic KOA — defined as clinically diagnosed knee osteoarthritis co-occurring with knee pain — imposes substantial mobility, quality-of-life, and health-care-utilization costs. Risk-prediction models may support earlier stratification, but publication-quality evidence requires more than high discrimination alone. Reporting should include calibration behaviour and threshold-level decision consequences, not only ROC-AUC, and should follow standards such as TRIPOD[2] and PROBAST[3].

A recent CHARLS-based report[1] presented strong discrimination for symptomatic KOA and highlighted biological age as a central predictor. However, two concerns arise when interpreting this finding. First, the “Biological Age” field supplied in the CHARLS dataset is pre-computed without documented methodology, raising questions about reproducibility and potential overfitting. Second, translating a BA-centric prediction model to other populations assumes that the biomarker–aging–disease relationship is biologically universal — an assumption that has not been tested across culturally distinct cohorts. This manuscript frames the project as an internal-validation study with a structured decomposition of the BA contribution and a cross-cultural replication on the RAND Health and Retirement Study (HRS) and the English Longitudinal Study of Ageing (ELSA) to characterise the transportability of the aging–KOA signal across populations.

2 Methods

2.1 Study Objective and Outcome

The prediction target was symptomatic KOA in a CHARLS-derived cohort. The outcome was operationalised from two CHARLS-derived fields as

$$\text{KOA} = 1[\text{Arthritis} = \text{“yes”}] \wedge 1[\text{position_knees} = \text{“yes”}],$$

which matches the “symptomatic KOA” operationalisation used by Fu et al.[1] (physician-reported arthritis with knee-localised joint pain). Both source fields were removed from the predictor set to prevent target leakage; rows with `position_knees` missing were dropped because the target itself could not be verified for those subjects.

2.2 Study Cohort and Exclusion Cascade

The analysis was built on Sheet1 of the CHARLS extract (15,545 baseline rows, 28 columns). Applying the target-verification rule removed 3,216 rows with `position_knees` missing, yielding the analytical cohort “Dataset A” with 12,329 rows and 13.32% symptomatic-KOA prevalence. Dataset A was then partitioned once, before any model development, into an internal training partition (80%, $n = 9,863$, prevalence 13.32%) and an unseen holdout partition (20%, $n = 2,466$, prevalence 13.30%) using a stratified split with `random_state=42`. Table 1 summarises the characteristics of Dataset A and its two partitions.

2.3 Feature Set (`raw17`)

The pre-specified primary feature set, referred to throughout as *raw17*, contains 17 variables (Table 2). The subset excluding Biological Age, wave, and Time is labelled *raw16* and is used as the harmonised

Table 1: Cohort characteristics of the CHARLS-derived Dataset A and the early 80/20 partitions. Values are counts (%) or mean $\pm$ SD.

Characteristic	Full Dataset A	Internal train (80%)	Unseen holdout (20%)
N	12,329	9,863	2,466
Symptomatic KOA, n (%)	1,642 (13.32)	1,314 (13.32)	328 (13.30)
Biological Age, mean $\pm$ SD	58.70 $\pm$ 9.79	58.70 $\pm$ 9.76	58.70 $\pm$ 9.90
Age bucket $\geq$ 60, n (%)	3,357 (27.2)	2,676 (27.1)	681 (27.6)
Female, n (%)	6,443 (52.3)	5,143 (52.1)	1,300 (52.7)
BMI, mean $\pm$ SD	23.65 $\pm$ 3.84	23.63 $\pm$ 3.83	23.73 $\pm$ 3.88
BMI overweight, n (%)	10,122 (82.1)	8,081 (81.9)	2,041 (82.8)
BMI obese, n (%)	1,488 (12.1)	1,192 (12.1)	296 (12.0)
Married, n (%)	10,429 (84.6)	8,343 (84.6)	2,086 (84.6)
Education low/medium/high, %	28.1 / 61.9 / 10.0	27.9 / 62.0 / 10.1	28.8 / 61.4 / 9.7
Residence urban, n (%)	7,954 (64.5)	6,388 (64.8)	1,566 (63.5)
Hypertension, n (%)	3,142 (25.5)	2,503 (25.4)	639 (25.9)
Dyslipidemia, n (%)	1,108 (9.0)	882 (8.9)	226 (9.2)
Diabetes, n (%)	726 (5.9)	584 (5.9)	142 (5.8)
Cancer, n (%)	93 (0.8)	75 (0.8)	18 (0.7)
CVD, n (%)	1,304 (10.6)	1,016 (10.3)	288 (11.7)
Smoking ever, n (%)	4,896 (39.7)	3,897 (39.5)	999 (40.5)
Drinking ever, n (%)	5,144 (41.7)	4,156 (42.1)	988 (40.1)

feature set for external validation.

Table 2: The 17 variables in the raw17 feature set used by the primary model.

Variable	Type	CHARLS-derived meaning
wave	integer	Survey wave identifier
Time	integer	Survey year indicator
Gender	binary	Self-reported sex (1=male, 2=female)
Age_New	ordinal (2)	Age bucket (45–59, $\geq$ 60)
Marital	binary	Marital status (married vs other)
Education	ordinal (3)	Educational attainment (low/medium/high)
Residence	binary	Urban vs rural
Hypertension	binary	Self-reported physician-diagnosed
Dyslipidemia	binary	Self-reported physician-diagnosed
Diabetes	binary	Self-reported physician-diagnosed
Cancer	binary	Self-reported physician-diagnosed
CVD	binary	Self-reported cardiovascular disease
Smoke	binary	Ever vs never smoker
Drink	binary	Ever vs never drinker
BMI	continuous	Measured body-mass index (kg/m^2)
BMI_New	ordinal (3)	BMI category (underweight/normal, overweight, obese)
Biological Age	continuous	CHARLS-supplied biological-age field

2.4 Missing Data Handling

After the target-verification exclusion, the raw17 feature set and the outcome were complete for every row in Dataset A (0% missingness). As a pipeline safety net and to support sensitivity branches with additional biomarker features, missing values were handled with train-fitted simple imputation (median for numeric, most-frequent for categorical). Imputers and standardisation (`StandardScaler`) were fit on training data only and then applied to validation or unseen data.

2.5 Early Unseen Holdout Strategy

The early 80/20 stratified split was created before model development and persisted as CSV artefacts (`dataset_A_internal_train.csv`, `dataset_A_unseen_test.csv`). Internal model development and OOF estimation used only the internal partition; the unseen partition was reserved for a single-shot external-style evaluation.

2.6 Primary and Secondary Modeling Strategy

The pre-specified primary model was a Random Forest (`RandomForestClassifier`, `n_estimators=300`, `random_state=42`, all other hyperparameters at scikit-learn defaults). No grid search, Bayesian search, or Optuna-based tuning was performed on this configuration. The BA replacement experiment (Section 2.12) additionally evaluated XGBoost and LightGBM for robustness.

2.7 Class-Imbalance Handling

No resampling (SMOTE, random under-/over-sampling) and no `class_weight` adjustment were applied to the primary Random Forest. PR-AUC was pre-specified as the primary discrimination metric alongside ROC-AUC, following standard recommendations for imbalanced binary prediction.

2.8 Evaluation Framework

Internal validation used stratified 5-fold out-of-fold (OOF) estimation on the internal training partition, followed by re-evaluation on the unseen holdout. Reported metrics:

- discrimination: ROC-AUC, PR-AUC;
- calibration: Brier score, ECE, MCE, Cox logistic calibration slope and intercept, decile reliability diagram;
- decision relevance: Vickers net benefit at thresholds 0.05–0.60 and threshold-band summaries for 0.10–0.30.

Probability calibration was assessed for both uncalibrated (“raw”) and isotonic-calibrated (`CalibratedClassifierCV` `method=“isotonic”`, 3-fold inner CV) variants.

2.9 Interpretability

Model interpretation included SHAP global importance[10], SHAP interaction analysis, and local waterfall plots for representative cases.

2.10 Differences vs. Reference Study (Fu et al., 2025)

Table 3 summarises the key methodological differences between the reference study and the present pipeline.

2.11 Biological Age Computation: KDM-BA and PhenoAge

The CHARLS dataset contains a pre-computed “Biological Age” field whose derivation method is not documented in the source publication[1]. To test whether this feature is reproducible using

Table 3: Methodological delta versus reference study.

Domain	Reference paper	Current pipeline
Cohort framing	Symptomatic KOA in CHARLS-derived sample	Symptomatic KOA in CHARLS-derived sample; updated internal protocol with early holdout
Missing data	Study-specific exclusions / imputation choices	Train-fitted <code>SimpleImputer</code> within each split
Split protocol	70/30 emphasised	Early 80/20 unseen holdout + internal OOF validation
Class-balance handling	Model-specific handling	No resampling, no <code>class_weight</code> adjustment
Primary metric	ROC-AUC-centred	PR-AUC reported alongside ROC-AUC
Primary model	XGBoost-centred	Random Forest + raw17 pre-specified

transparent methods, we computed two alternative biological age constructs from 8 blood biomarkers available in CHARLS: platelet count, C-reactive protein (CRP), HbA1c, creatinine, blood urea nitrogen (BUN), total cholesterol, triglycerides, and systolic blood pressure (SBP).

KDM-BA. The Klemes-Doubal Method[11] computes BA as a weighted combination of biomarker regressions on chronological age. For each biomarker x_i , a linear regression $x_i = a_i + b_i \cdot \text{CA} + \epsilon_i$ is fitted, yielding intercept a_i , slope b_i , and residual standard error σ_i . The KDM biological age is:

$$\text{BA}_{\text{KDM}} = \frac{\sum_i \frac{(x_i - a_i) b_i}{\sigma_i^2} + \frac{\text{CA}}{\sigma_{\text{CA}}^2}}{\sum_i \frac{b_i^2}{\sigma_i^2} + \frac{1}{\sigma_{\text{CA}}^2}}$$

We computed KDM-BA from both log-transformed (`BA_KDM_log`) and original-scale (`BA_KDM_orig`) biomarker values. The “Biological Age” column in Sheet1 was used as the chronological age proxy, as CHARLS does not provide a continuous age variable (`Age_New` is a binary ≥ 60 indicator).

PhenoAge_Adapted. Because canonical Levine PhenoAge[12] requires biomarkers not available in CHARLS (e.g., albumin, lymphocyte count), we constructed an adapted phenotypic-age surrogate using winsorised z-scores of the same 8 biomarkers, weighted by clinical plausibility (CRP: +1.40, HbA1c: +1.20, creatinine: +1.00, BUN: +0.90, SBP: +1.00, total cholesterol: +0.60, triglycerides: +0.60, platelet: −0.40) and mapped to age units via a scaling transformation. This score is explicitly non-canonical and is labelled *PhenoAge_Adapted* throughout to avoid misinterpretation.

Biological Illness Risk (BIR). BIR was defined as $\text{BA}_{\text{KDM}} - \text{CA}$, capturing the acceleration or deceleration of biological relative to chronological age.

2.12 Biological Age Replacement Experiment

To determine whether the pre-computed Sheet1 BA is irreplaceable or whether any transparent biomarker-derived BA achieves equivalent performance, we conducted a controlled replacement experiment (Step 18). Six configurations were evaluated using the same cross-validation protocol as the feature impact analysis (3 seeds $\times$ 5-fold StratifiedKFold, Random Forest with StandardScaler):

1. **raw17 (Sheet1 BA)** — baseline, 17 features with the original BA field.

2. **raw16_only (no BA)** — 16 features, BA removed entirely.
3. **raw16 + KDM_log** — KDM-BA (log-transformed) replaces Sheet1 BA, 17 features.
4. **raw16 + KDM_orig** — KDM-BA (original scale) replaces Sheet1 BA, 17 features.
5. **raw16 + PhenoAge_Adapted** — adapted PhenoAge replaces Sheet1 BA, 17 features.
6. **raw16 + PhenoAge_Accel** — PhenoAge acceleration replaces Sheet1 BA, 17 features.

All configurations use the same $n = 12,329$ CHARLS Dataset A and the same evaluation harness. XGBoost (200 trees, max depth 6) and LightGBM (200 trees) were additionally evaluated for robustness.

2.13 ML-Predicted Biological Age for External Transport

Because ELSA and HRS lack the biomarkers needed to compute KDM-BA or PhenoAge directly, we tested whether a demographic-predicted BA could transport the aging signal. Two ML regressor targets were trained on CHARLS internal training data using only raw16 features (the sociodemographic variables available in all three cohorts):

1. **ML_BA** — predict the Sheet1 “Biological Age” column (CV $R^2 = 0.636$, holdout $R^2 = 0.633$, MAE = 4.7 years).
2. **ML_KDM_BA** — predict the computed KDM-BA (CV $R^2 = 0.482$, holdout $R^2 = 0.501$, MAE = 6.0 years).

Each target was predicted by an ensemble of Random Forest, XGBoost, and LightGBM regressors (300 trees each). The fitted models were then applied to the harmonised ELSA and HRS raw16 data to generate predicted BA columns, which were added to the raw16 feature set for KOA prediction. Four scenarios per cohort were evaluated:

1. raw16 only (no BA — baseline)
2. raw16 + ML_BA
3. raw16 + ML_KDM_BA
4. raw16 + ML_BA + ML_KDM_BA (ensemble BA — addresses the ensemble prediction option)

2.14 AUC Gap Decomposition Strategy

To disentangle the feature-subset penalty (attributable to BA absence) from the cross-cultural penalty (attributable to population differences), we conducted the transportability analysis in three stages:

1. **CHARLS raw16 internal baseline.** The primary model was re-trained and evaluated on the CHARLS internal partition using only the features common to CHARLS, ELSA, and HRS (raw16). Because both training and evaluation occur within CHARLS, this isolates the feature-subset penalty with no cross-cultural confound.

2. **External application.** The raw16 model trained on CHARLS was applied to ELSA Wave 8 and HRS Wave 13 without re-fitting. The gap between CHARLS raw16 holdout performance and external cohort performance is the cross-cultural attenuation, net of the feature-subset penalty.
3. **Reduced Aging Score (RAS) sensitivity.** To test whether a harmonised biological age proxy could recover the feature-subset gap, we constructed RAS from four biomarkers available in both CHARLS and ELSA (hs-CRP, HbA1c, systolic BP, total cholesterol). RAS was trained as a linear predictor of CHARLS KDM-BA ($R^2 = 0.417$) and added to the raw16 model for ELSA external validation.

2.15 Reduced Aging Score (RAS)

RAS was constructed as a linear combination of four biomarkers:

$$\text{RAS} = \beta_0 + \beta_1 \cdot \ln(\text{CRP}) + \beta_2 \cdot \ln(\text{HbA1c}) + \beta_3 \cdot \ln(\text{TC}) + \beta_4 \cdot \text{SBP}$$

The coefficients $\beta_{0..4}$ were estimated by ordinary least squares regression of CHARLS KDM-BA on the four CHARLS biomarker equivalents (`crp_mg.L`, `Hb_A1c`, `TC_mg.d.L`, `sbp.mean`). R^2 on CHARLS data was 0.417 ($n = 12,329$), indicating that RAS captures 41.7% of KDM-BA’s internal variance.

For ELSA, biomarkers were converted to CHARLS-equivalent units: hs-CRP (mg/L) log-transformed directly; HbA1c converted from IFCC (mmol/mol) to DCCT (%) via $\text{HbA1c\%} = (\text{mmol/mol}/10.929) + 2.15$, then log-transformed; total cholesterol converted from mmol/L to mg/dL ($\times 38.67$), then log-transformed; systolic BP used directly (mmHg). After excluding ELSA respondents with missing biomarker codes ($-8, -2, -1$), 2,499 of 8,416 respondents had complete biomarker data for RAS computation.

2.16 HRS and ELSA Cross-Cultural Replication

HRS Cohort and Data Source

The Health and Retirement Study (HRS)[13] is a US nationally representative longitudinal panel of adults aged ≥ 50 . For this replication we used the RAND HRS Longitudinal File 2022 (Version 1)[14]. Wave 13 (survey year 2016) was selected as the target wave.

HRS Eligibility and KOA Proxy Definition

Inclusion criteria: respondents with Wave 13 interview data and age ≥ 45 years ($n = 20,640$ after age filter). A two-component KOA proxy was constructed: (1) current physician-reported arthritis (`r13arthr` = 1) and (2) knee-relevant functional impairment (`r13stoopa` = 1). This proxy results in a prevalence of 33.9% ($n_{\text{pos}} = 6,976$). Rows with missing arthritis response were excluded ($n = 35$), yielding a final analytic sample of $n = 20,605$.

ELSA Cohort and Data Source

The English Longitudinal Study of Ageing (ELSA) is a UK nationally representative longitudinal panel of adults aged ≥ 50 . We used the Gateway Harmonized ELSA file (Version H, 2002–2023), distributed through the UK Data Service (SN 5050). Wave 8 (2016–2017) was the primary target wave; Wave 9 (2018–2019) was retained as a temporal sensitivity analysis.

ELSA Eligibility and KOA Proxy Definition

Inclusion criteria: Wave 8 interview data and age ≥ 45 years ($n = 8,416$ after age filter). A two-component KOA proxy was constructed: (1) ever-diagnosed arthritis (`r8arthre = 1`) and (2) difficulty stooping/kneeling/crouching (`r8stoopa = 1`). This proxy results in a prevalence of 24.9% ($n_{\text{pos}} = 2,097$).

External Feature Harmonisation (raw16)

Both HRS and ELSA were harmonised to the raw16 feature set (raw17 minus Biological Age, wave, and Time). HRS yielded 12 of 17 raw17 features (Biological Age, Residence, Dyslipidemia, and wave excluded); ELSA yielded 14 of 17 (Biological Age, wave, and Time excluded; Residence and Dyslipidemia available). The trained CHARLS model was re-fitted on the raw16 subset of the CHARLS internal training partition and then applied to each external cohort.

3 Results

3.1 Primary Model Performance (Random Forest + raw17)

Table 4 summarises the matched internal OOF and unseen-holdout performance for the pre-specified primary model.

Table 4: Primary model (Random Forest + raw17) metrics.

Set	Variant	ROC-AUC	PR-AUC	Brier	ECE	Slope	Intercept	F1
Internal OOF	Raw	0.847	0.688	0.070	0.045	0.989	-0.040	0.649
Internal OOF	Isotonic	0.839	0.676	0.069	0.042	1.607	1.000	0.653
Unseen holdout	Raw	0.907	0.791	0.053	0.056	1.216	0.178	0.801
Unseen holdout	Isotonic	0.900	0.779	0.051	0.060	1.742	1.088	0.799

3.2 Decision-Curve Analysis

Across the 10%–30% threshold band, the primary model produced strictly positive mean net benefit against both treat-all and treat-none. The derived net-reduction-per-100-patients figure[6] was 56.0 (internal OOF) and 63.8 (unseen holdout) avoided unnecessary interventions versus treat-all; for isotonic probabilities the corresponding values were 56.5 and 65.8, with relative reductions of 65.2% and 75.9% against the treat-all baseline.

3.3 Biological Age Replacement Experiment

Table 5 presents the BA replacement experiment. Replacing Sheet1 BA with any transparent alternative degraded performance; removing BA entirely performed better than replacing it with KDM-BA.

Three observations emerge. First, Sheet1 BA is the best BA variant in the full model (0.896), but its contribution is modest (-0.037 when removed). Second, replacing BA with any transparent alternative *actively hurts*: raw16 + KDM_orig (0.831) is worse than raw16_no_BA (0.859), indicating multicollinearity noise from inserting a redundant biomarker-derived feature. Third, the maximum penalty from any BA manipulation (method replacement or removal) is 0.067 AUROC — small relative to the 0.26 cross-cultural gap reported below.

Table 5: Biological Age replacement experiment (Random Forest, 3 seeds $\times$ 5-fold CV, $n = 12,329$). All configurations preserve the same 17-feature count except raw16_only.

Configuration	Features	ROC-AUC	PR-AUC	Brier	Δ from raw17
raw17 (Sheet1 BA)	17	0.896	0.776	0.057	— (baseline)
raw16_only (no BA)	16	0.859	0.578	0.071	−0.037
raw16 + PhenoAge_Adapted	17	0.841	0.561	0.083	−0.055
raw16 + KDM_orig	17	0.831	0.525	0.086	−0.065
raw16 + KDM_log	17	0.829	0.513	0.087	−0.067
raw16 + PhenoAge_Accel	17	0.798	0.444	0.094	−0.098

3.4 Single-Feature BA Predictive Value Across Cohorts

Table 6 presents each BA variant as a standalone KOA predictor (univariate logistic regression). This directly addresses the question: does BA carry KOA signal on its own, and does that signal transfer?

Table 6: Single-feature BA $\rightarrow$ KOA AUROC across all cohorts. “—” indicates the BA variant cannot be computed (missing biomarkers for KDM/PhenoAge; Sheet1 BA only in CHARLS). ML-predicted BA is a demographic proxy (Step 16/17).

Feature	CHARLS ($n=12,329$)	ELSA ($n=8,416$)	HRS ($n=20,605$)
Biological Age (Sheet1)	0.536	—	—
KDM-BA (log)	0.516	—	—
KDM-BA (orig)	0.519	—	—
PhenoAge_Adapted	0.530	—	—
PhenoAge_Accel	0.504	—	—
BIR_log	0.535	—	—
ML_BA (predicted)	—	0.556	0.629
ML_KDM_BA (predicted)	—	0.589	0.629
Age_New (no BA, baseline)	—	0.553	0.599

All BA variants in CHARLS achieve only 0.50–0.54 AUROC — barely above chance. The best (Sheet1 BA at 0.536) has a KOA correlation of $r = 0.038$. This refutes the interpretation that BA is a strong standalone KOA predictor: its apparent importance in the full raw17 model (SHAP = 0.439) reflects conditional interaction effects, not marginal predictive power.

In HRS, ML-predicted BA achieves 0.629 — but Age_New alone achieves 0.599 (HRS has 33.9% KOA prevalence vs. CHARLS’s 13.3%). The BA increment over age is only +0.030. In ELSA, ML_BA (0.556) is nearly identical to Age_New (0.553). ML_KDM_BA consistently adds more over age than ML_BA (+0.036 vs. +0.003 in ELSA; +0.030 vs. +0.030 in HRS), suggesting that the biomarker-derived KDM target encodes more KOA-relevant signal than the undocumented Sheet1 BA formula.

3.5 ML-Predicted BA Applied to External Cohorts

Table 7 presents the effect of adding ML-predicted BA to the raw16 model in ELSA and HRS.

The ensemble BA scenario (raw16 + ML_BA + ML_KDM_BA) achieved the highest AUROC in ELSA (0.613, +0.009 over raw16) but not in HRS (0.607). ML_KDM_BA alone was best in HRS (0.610, +0.006). All gains are within ± 0.01 of raw16 — negligible relative to the 0.26 cross-cultural gap.

Table 7: ML-predicted BA applied to external cohorts. All scenarios use the CHARLS-trained Random Forest re-fitted on raw16. ML_BA and ML_KDM_BA are demographic-predicted BA from Step 16. The ensemble scenario (raw16 + ML_BA + ML_KDM_BA) addresses the ensemble BA prediction option.

Cohort	Scenario	n	ROC-AUC	PR-AUC	Brier	F1
ELSA	raw16	8,416	0.604	0.297	0.201	0.427
ELSA	raw16 + ML_BA	8,416	0.602	0.307	0.191	0.422
ELSA	raw16 + ML_KDM_BA	8,416	0.597	0.300	0.192	0.421
ELSA	raw16 + ML_BA + ML_KDM_BA	8,416	0.613	0.319	0.187	0.425
HRS	raw16	20,605	0.604	0.399	0.258	0.525
HRS	raw16 + ML_BA	20,605	0.607	0.410	0.250	0.526
HRS	raw16 + ML_KDM_BA	20,605	0.610	0.412	0.248	0.525
HRS	raw16 + ML_BA + ML_KDM_BA	20,605	0.607	0.412	0.247	0.523

3.6 AUC Gap Decomposition: Feature-Subset vs. Cross-Cultural

Table 8 presents the structured decomposition of the 0.303 ROC-AUC gap between the CHARLS raw17 holdout and the external cohorts.

Table 8: AUC gap decomposition across CHARLS raw17, CHARLS raw16, ELSA, and HRS. All values are raw (uncalibrated) ROC-AUC except where noted.

Stage	Cohort	Features	n	KOA%	ROC-AUC	Δ from baseline
A	CHARLS raw17 Holdout	17	2,466	13.3	0.907	— (baseline)
B	CHARLS raw16 Holdout	15	2,466	13.3	0.869	−0.038 (feature-subset)
C	ELSA W8 raw16	14	8,416	24.9	0.604	−0.303 (total)
C'	HRS W13 raw16	12	20,605	33.9	0.604	−0.303 (total)
Feature-subset component						−0.038 (13%)
Cross-cultural component						−0.265 (87%)

Feature-subset penalty = B − A. Cross-cultural penalty = C − B. Percentages are of the total A→C gap (0.303).

The feature-subset penalty (dropping Biological Age, wave, and Time from raw17) accounts for 13% of the total AUC loss. The remaining 87% is cross-cultural attenuation occurring between the CHARLS and external cohorts.

Notably, the ELSA and HRS raw16 ROC-AUC values are nearly identical (0.604) despite ELSA having two more harmonised features (Residence and Dyslipidemia), using a different KOA proxy (ever-diagnosed vs. current-wave arthritis), and drawing from a different population (UK vs. US). This convergence supports a stable transportability ceiling around AUROC $\approx$ 0.60 for raw16 without Biological Age across culturally distinct aging cohorts.

3.7 Reduced Aging Score (RAS) Experiment

Table 9 presents the effect of adding RAS to the ELSA raw16 model.

Adding RAS — the best biological age proxy constructible from publicly available ELSA biomarkers — recovers only 0.006 ROC-AUC, or approximately 2% of the total cross-cultural gap.

3.8 ELSA Wave 9 Sensitivity Analysis

A temporal sensitivity analysis using ELSA Wave 9 (2018–2019, $n = 8,683$, KOA proxy prevalence 23.3%) produced consistent results: ROC-AUC 0.596 (raw), calibration slope 0.187. After split-

Table 9: Effect of the Reduced Aging Score on ELSA external validation. RAS captures 41.7% of KDM-BA variance in CHARLS.

Model	ROC-AUC	PR-AUC	Brier
ELSA raw16	0.6050	0.297	0.200
ELSA raw16 + RAS	0.6111	0.316	0.188
Δ (RAS gain)	+0.0061	+0.019	−0.013

calibrate-evaluate isotonic recalibration: ROC-AUC 0.615, slope 1.225, ECE 0.009. Wave 9 lacks a harmonized BMI variable, so BMI was fully imputed from CHARLS training data. The results confirm temporal robustness.

4 Discussion

Internal validation. The primary raw17 model achieved ROC-AUC 0.847 (OOF) / 0.907 (holdout) and PR-AUC 0.688 / 0.791, substantially exceeding the prevalence baseline. Strictly positive DCA net benefit in the 10%–30% threshold band provides a grounded internal-evidence basis for risk-stratification decision support.

The BA contribution is modest and method-dependent. The BA replacement experiment (Table 5) reveals that Sheet1 BA contributes 0.037 AUROC to the raw17 model. Replacing it with transparent KDM-BA or PhenoAge degrades performance further (−0.055 to −0.098), not because these methods are inferior BA estimators, but because inserting a redundant biomarker-derived feature into raw16 introduces multicollinearity noise. The single-feature analysis (Table 6) confirms that all BA variants — including Sheet1’s — are weak standalone KOA predictors (0.50–0.54 AUROC, $r = 0.02$ –0.04). BA’s apparent importance in the full model (SHAP = 0.439) reflects conditional interaction effects, not marginal predictive value.

The biomarker–BA–KOA relationship is population-specific. The central hypothesis tested in this study is that the aging–disease relationship established in CHARLS does not transfer to other populations. Five lines of evidence converge:

1. **BA is weak even in CHARLS.** All BA variants achieve only 0.50–0.54 AUROC as single-feature KOA predictors ($r = 0.02$ –0.04). If BA were a strong biological signal for KOA, one would expect higher standalone discrimination.
2. **Removing BA costs only 0.037 AUROC.** The BA removal penalty is a small fraction of the total 0.303 cross-cohort gap.
3. **ML-predicted BA adds negligible signal externally.** Demographic-predicted BA (ML_BA, ML_KDM_BA) improves raw16 by +0.003 to +0.010 AUROC in ELSA/HRS. The ensemble BA scenario adds +0.009. These gains are at least 25× smaller than the cross-cultural gap.
4. **ELSA–HRS convergence despite differences.** ELSA and HRS produce near-identical ROC-AUC (0.604) despite different feature availability (14 vs. 12 variables), KOA proxy definitions (ever-diagnosed vs. current-wave arthritis), and populations (UK vs. US). This convergence is not compatible with a BA-centric narrative: if missing BA were the primary driver, cohorts with different feature completeness should perform differently.

5. **RAS recovers negligible signal.** The best biological age proxy constructible from ELSA biomarkers recovers $+0.006$ ROC-AUC — less than 2% of the cross-cultural gap.

AUC gap decomposition: cross-cultural dominates. The structured decomposition (Table 8) shows that 87% of the total AUC gap is cross-cultural, not feature-subset. The complete evidence chain is:

Table 10: Evidence chain: BA method choice and availability produce small penalties; the cross-cultural cliff dominates.

Stage	Description	ROC-AUC	Δ
1	raw17 + Sheet1 BA (CHARLS)	0.896	—
2	raw16 + KDM-BA (CHARLS)	0.831	-0.065 (BA method)
3	raw16 no BA (CHARLS)	0.859	-0.037 (BA removal)
4	raw16 no BA (ELSA)	0.604	-0.292 (cross-cultural)
4	raw16 no BA (HRS)	0.604	-0.292 (cross-cultural)
5	raw16 + ML-BA ensemble (ELSA)	0.613	$+0.009$ (ML-BA recovers)
5	raw16 + ML-KDM-BA (HRS)	0.610	$+0.006$ (ML-BA recovers)

No manipulation of BA — using the original formula, replacing it with KDM-BA, removing it entirely, or predicting it from demographics — can account for the 0.26 cross-cultural gap. The BA-related penalties (0.037–0.067) are 4–7 $\times$ smaller than the cross-cultural penalty (0.265). Even if full biomarker-derived BA were available in ELSA/HRS, the expected recovery is bounded by what BA contributes in CHARLS (~ 0.04 AUROC), which is $< 15\%$ of the gap.

Comparison with the reference study. The reference study by Fu et al.[1] reported XGBoost-centred discrimination with biological age as the dominant predictor (SHAP > 0.6), using a 70/30 split. The present pipeline replicates the same outcome operationalisation and confirms that the raw17 feature set supports strong discrimination. However, our analysis adds three findings that qualify the original interpretation: (1) the pre-computed BA field has undocumented methodology and is only moderately better than transparent KDM-BA (0.896 vs. 0.831); (2) BA is a weak standalone predictor (0.536 AUROC, $r = 0.038$) whose apparent importance reflects interaction effects; and (3) the BA–KOA relationship does not transport across populations.

Strengths. Strengths include the large population-based cohort, transparent leakage controls, pre-specification of the primary model, multi-dimensional evaluation (discrimination, calibration, clinical utility, interpretability), adherence to TRIPOD and PROBAST, the BA replacement experiment directly testing whether Sheet1 BA is reproducible, the ML-BA transport experiment, and the structured AUC gap decomposition.

5 Limitations

- **Outcome proxy heterogeneity is the primary limitation.** The AUC gap decomposition shows that 87% of performance loss is cross-cultural, with outcome definition mismatch (CHARLS knee-localised pain vs. HRS/ELSA stooping difficulty) likely the largest single contributor. Until a harmonised KOA definition is applied across all three cohorts, the true transportability of the raw17 model cannot be estimated.
- **Sheet1 Biological Age is undocumented.** The pre-computed BA field in the CHARLS dataset lacks published methodology. Our analysis shows its residual (after removing age-

group variance) correlates strongly with biomarkers (SBP $r = 0.41$, creatinine $r = 0.32$, BUN $r = 0.32$), suggesting it was derived from the same biomarkers used in our KDM-BA computation. However, the exact formula — and therefore its susceptibility to overfitting or information leakage — cannot be verified.

- **ML-predicted BA is a demographic proxy, not real BA.** The ML-BA models predict BA from sociodemographic features ($R^2 = 0.64$ for Sheet1 BA, 0.48 for KDM-BA). They capture only the demographic-BA relationship, not the biomarker-BA relationship. A reviewer could argue that real biomarker-derived BA in ELSA/HRS would perform better. However, given that BA contributes only 0.037 AUROC in CHARLS and is a weak standalone predictor (0.536), the expected gain is bounded and cannot account for the 0.26 gap.
- **ELSA and HRS lack full biomarker panels.** KDM-BA cannot be computed in ELSA (missing creatinine, BUN, platelets) or HRS (requires VBS linkage). RAS, based on 4 of 8 KDM biomarkers, captures only 42% of KDM-BA variance. We acknowledge that a cohort with complete biomarkers and the same KOA definition as CHARLS would provide a fairer test. However, the near-identical ELSA/HRS performance (0.604 each) despite different feature availability suggests feature completeness is not the binding constraint.
- **Cross-sectional outcome.** The KOA label is cross-sectionally derived and cannot distinguish incident from prevalent cases.
- **Self-reported comorbidities.** Hypertension, diabetes, cancer, CVD, dyslipidemia, and lifestyle variables are self-reported in CHARLS, inheriting recall bias and diagnostic ascertainment heterogeneity.
- **Bootstrap confidence intervals not yet tabulated.** Point estimates and reliability diagrams are the current evidence basis for calibration; bootstrap CIs for discrimination and calibration metrics will be added in the next revision (TRIPOD item 16).

6 Conclusion

This study demonstrates a methodologically rigorous internal-validation pipeline for symptomatic KOA prediction in a large Chinese nationally-representative cohort. The pre-specified primary model (Random Forest + raw17) achieves strong discrimination (ROC-AUC 0.907 on the unseen holdout), acceptable calibration, and positive DCA net benefit in the clinically relevant 10%–30% risk threshold band.

A structured decomposition of the biological age contribution reveals that BA — whether pre-computed by an undocumented formula, transparently computed via KDM, or demographic-predicted via ML — is a weak standalone KOA predictor (0.50–0.54 AUROC) whose contribution to the full model is modest (0.037 AUROC) and whose relationship with KOA does not transfer across populations. ML-predicted BA applied to external cohorts adds +0.003 to +0.010 AUROC; an ensemble of two ML-BA variants adds +0.009. All BA-related effects combined are bounded at ~ 0.067 AUROC.

The cross-cultural gap (0.265 AUROC, 87% of the total) is the dominant barrier to transportability. The near-identical performance of the raw16 model on ELSA and HRS (0.604) despite different feature availability, KOA proxy definitions, and populations establishes a conservative transportability ceiling that is not explained by BA absence. These findings reframe the external validation challenge: the primary obstacle is not a missing feature but a non-transferable disease–aging relationship. Future work should prioritise (i) harmonised KOA outcome definitions across cohorts, (ii) prospective

validation in cohorts where the same KOA definition and full biomarker panels are available, and (iii) investigation of population-specific effect modifiers that drive the cross-cultural gap.

Data Availability

The CHARLS development dataset (“Raw Data of Biological Age” derived from CHARLS) is publicly available under a CC BY 4.0 licence at Mendeley Data: <https://data.mendeley.com/datasets/3rv7mf5pv9/1> (Fu F, Mendeley Data V1, 2025; doi:10.17632/3rv7mf5pv9.1). The RAND HRS Longitudinal File is publicly available at <https://hrsdata.isr.umich.edu/data-products/rand> upon free registration. The English Longitudinal Study of Ageing (ELSA) is available from the UK Data Service (SN 5050) at <https://beta.ukdataservice.ac.uk/datacatalogue/studies/study?id=5050> under an End User Licence. The complete analysis pipeline code is available at the project repository.

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